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Engineering

UofT Phishnet

A Solution for UofT Phishing Email Problems

MIE1517 Introduction to Deep Learning: Group 11 Final Presentation

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University of Toronto

PhishNet

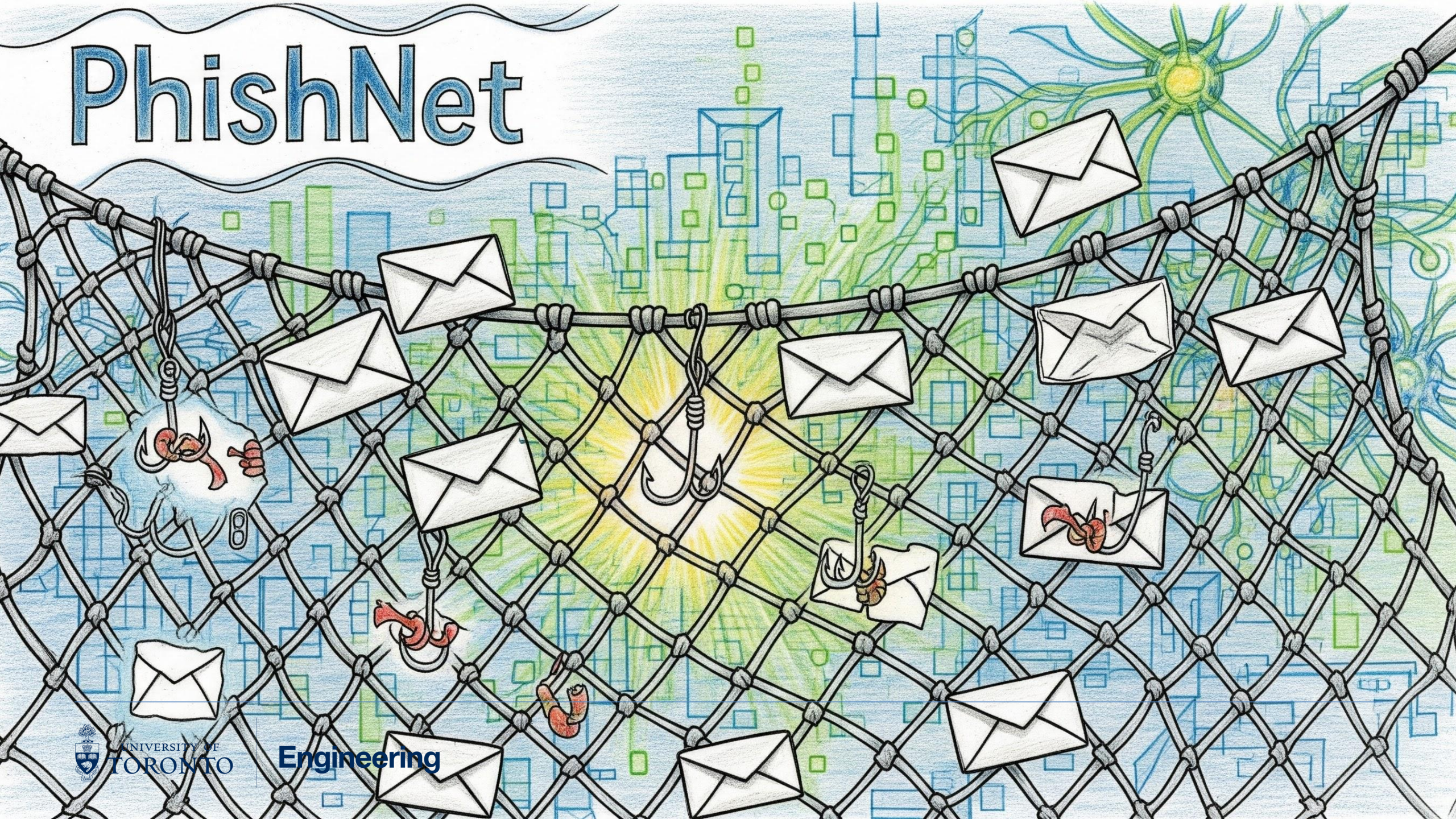


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Engineering

Introduction

- Motivational Background
- Problem Definition

Introduction

Motivational Background

 Eunjin Ko

  Reply  Reply all  Forward    

Mon 08/09/2025 17:58

 You replied on Mon 08/09/2025 19:32

Dear Student,

We are writing to remind you that our records indicate your **Fall 2025 tuition fees remain unpaid**. As per the **University of Toronto's Financial Aid & Registration Policy**, an **initial deposit of \$2,000–\$5,000 is required by today, September 8, 2025**, in order to secure your course registration and maintain uninterrupted access to academic services.

To complete your payment, please send an **Interac e-Transfer** to the University's BACPURSE account using the following details:

 **Email:** Nckamogu97@gmail.com 

 **Message/Memo:** Please include your student number

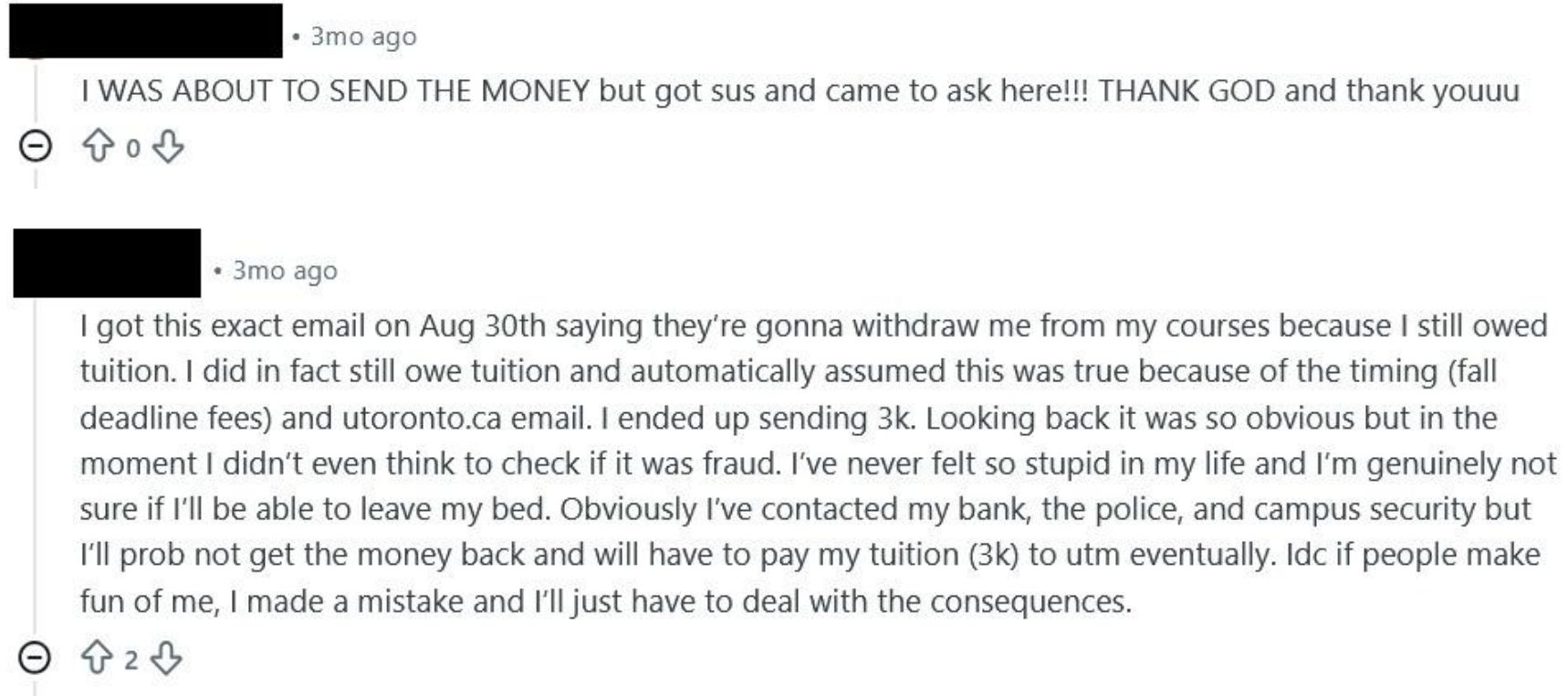
This initial deposit is **mandatory**, even if you have applied for **OSAP or are awaiting external funding**. OSAP disbursements are typically processed **after the term begins**, and therefore do not exempt students from the upfront deposit requirement.

Failure to submit the initial deposit today may result in:

- Removal from registered courses
- Holds on academic records (transcripts, graduation eligibility)

Introduction

Motivational Background



Introduction

Motivational Background



Associate Director, Information Security Strategic Initiatives

SEP 19



Marc Bishara • 9:35 AM

Hi [REDACTED],

Thank you for accepting my request. I got your contact from [REDACTED]. I am an MEng student enrolled in MIE. I am working along with three other colleagues (all PhD candidates) on a project for efficient classification of phishing emails as part of a deep learning class: MIE1517 with Prof Sinisa Colic. We were wondering if there could be room for collaboration with your department in our project. If that is something of interest to you, please let me know and I would be happy to share more details.

Best Regards
Marc Bishara
marc.bishara@mail.utoronto.ca



[REDACTED] • 9:50 AM

Hi Marc. We would be happy to help. What kind of collaboration are you looking for?



To: Marc Bishara

Cc: [REDACTED]



Reply



Reply all



Forward



Tue 18/11/2025 16:49



You replied on Tue 18/11/2025 17:24

Hi Marc,

Thank you for your email. Yes this request has been brought to my attention and I am looping back [REDACTED] in the email chain for visibility.

Please be advised we have reached out internally to the respective team to identify the level of data that is available for us to share and we shall reach back to you once we have more information on this.

Thanks

[REDACTED]
IS Risk Manager | Information Security| ITS

For General Risk related enquiries, please raise a ticket at [Get Help](#)



Dataset

Source	Email count
Arifa Islam, C. (2023)	203K over 11 datasets
Chakraborty, S. (2023)	18K
Miltchev, R. (2025)	2K
Radev, D. (2008)	11.5K
Hunter Biden dataset	A subset of 1K safe emails
Synthetic dataset	242

Total 235,742 from which we sampled 70,000 for training; 15,000 for validation and 15,000 for test

- Train: 70,000 | Validation: 15,000 | Test: 15,000

- ### Label Distribution Across Splits (0=Not Phishing, 1=Phishing)

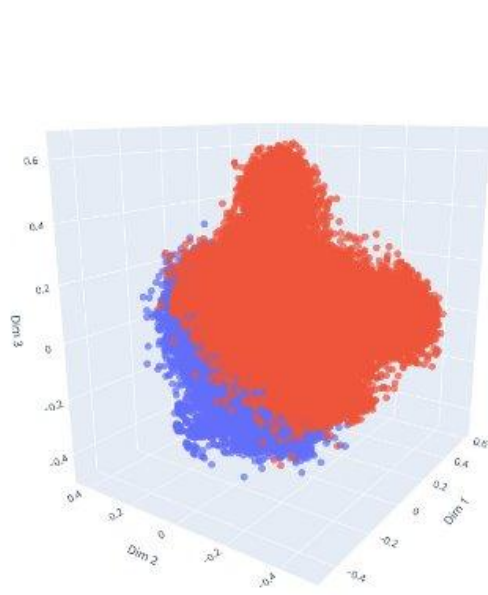


Train - Not Phishing (Label 0)

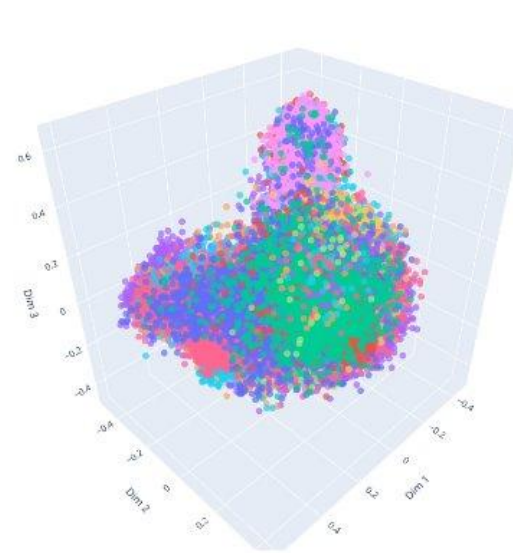


Dataset – PCA Analysis

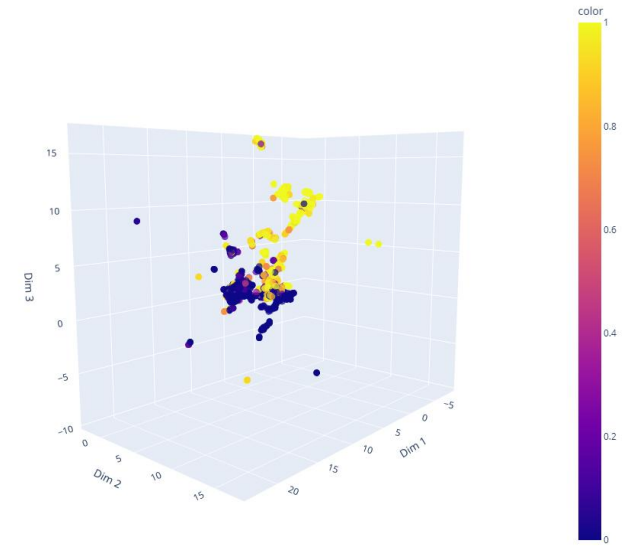
70K train emails Embedded with all-MiniLM-L6-v2 reduced with PCA colored by label



70K train emails Embedded with all-MiniLM-L6-v2 reduced with PCA colored by source



2404 clusters created with HDBSCAN and colored by risk score



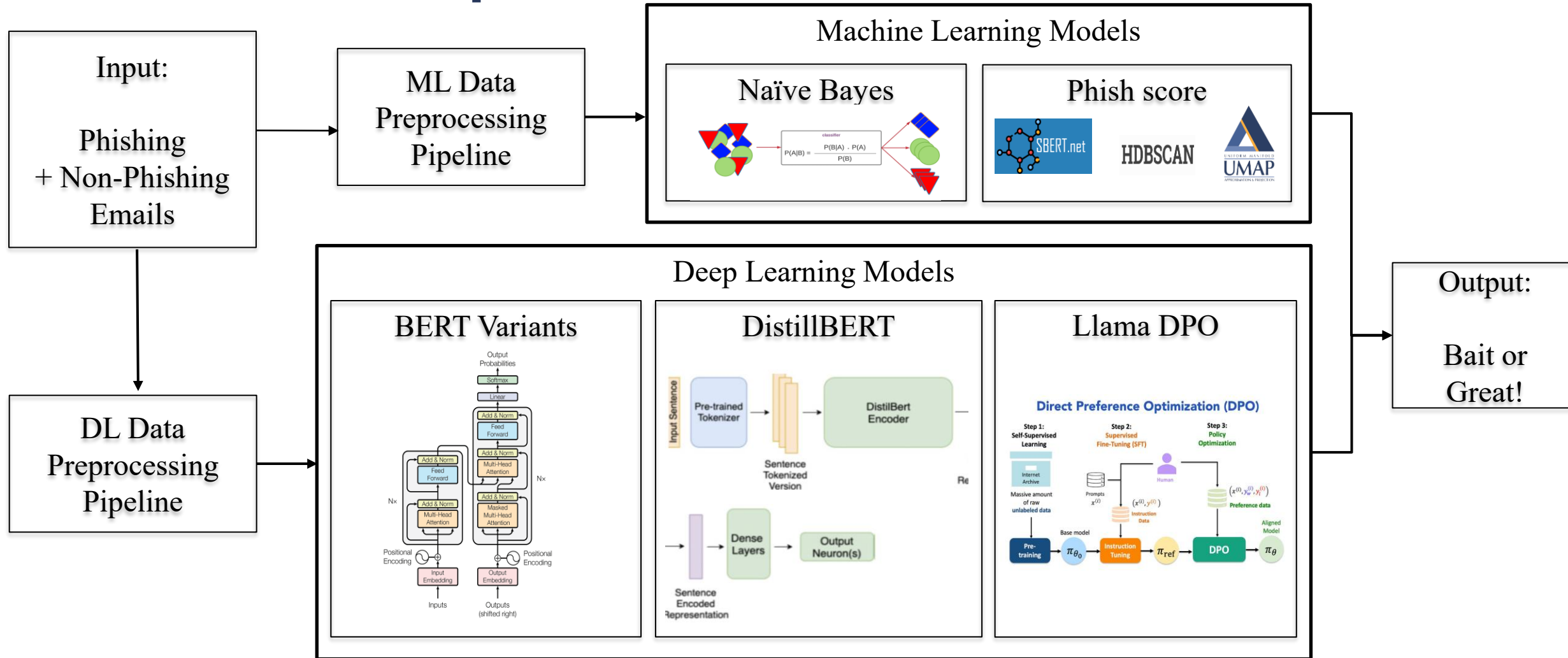


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End-to-End Pipeline

End-to-End Pipeline





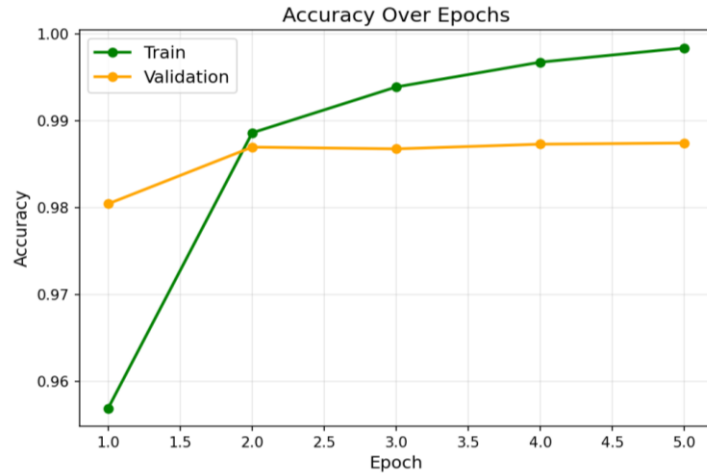
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Results

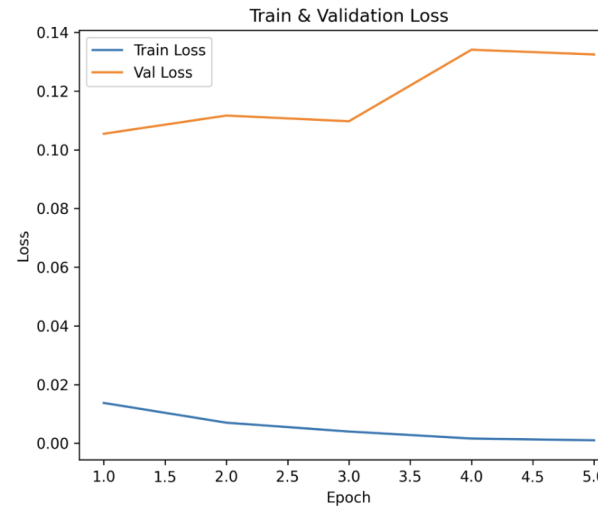
Results - Training

RoBERTa



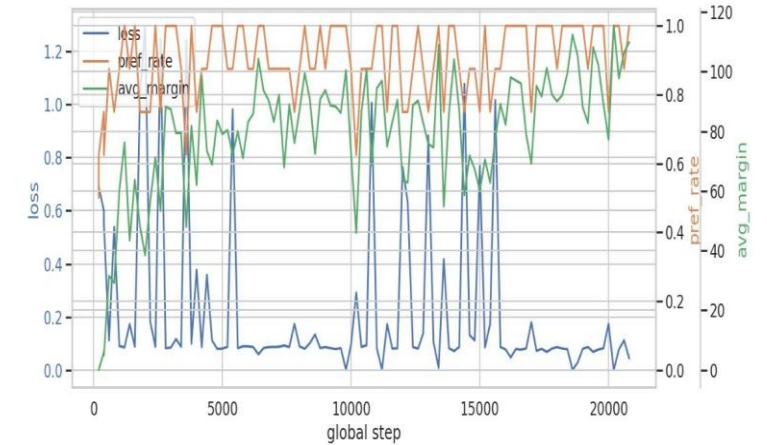
Training Epochs: 5
Best Validation Accuracy: 0.989

DistilBERT



Training Epochs: 5
Best Validation Accuracy: 0.983

Llama 3.2



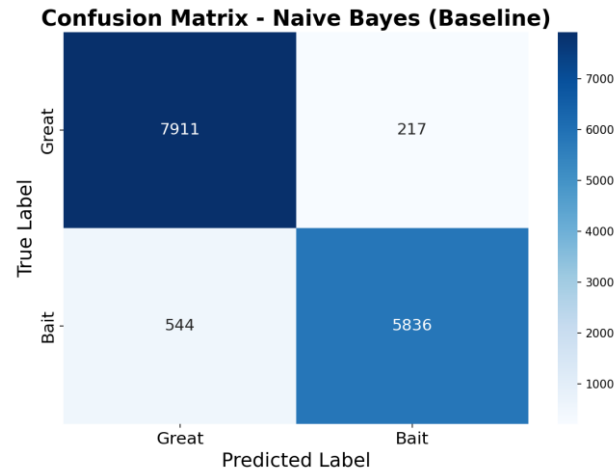
Training Epochs: 3
Best Validation Accuracy: 0.989

Results – Test Inference Results

Naïve Bayes Model

Accuracy: 94.75%

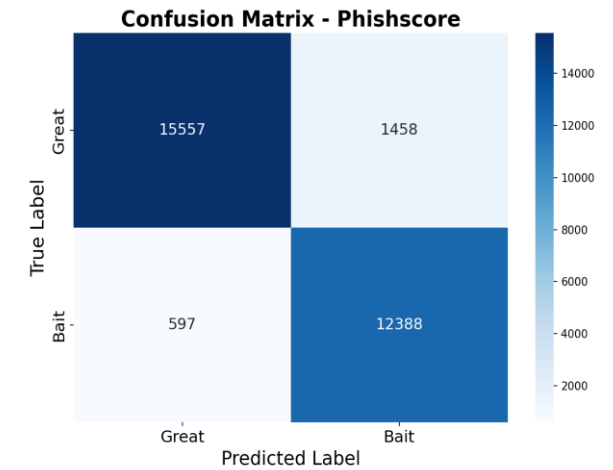
Weighted
Precision: 94.82%



Phish Score Model

Accuracy: 94.0%

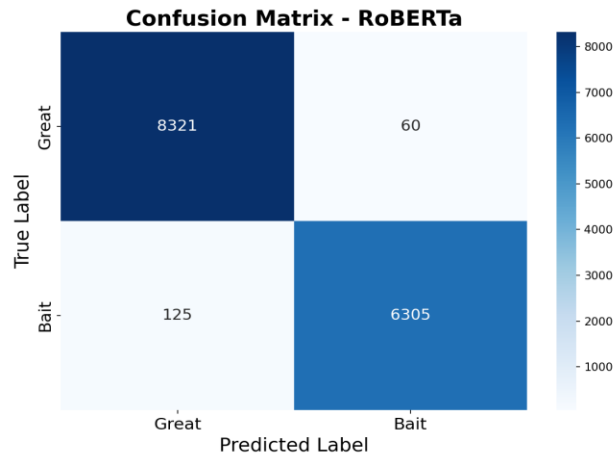
Weighted
Precision: 93.35%



RoBERTa Model

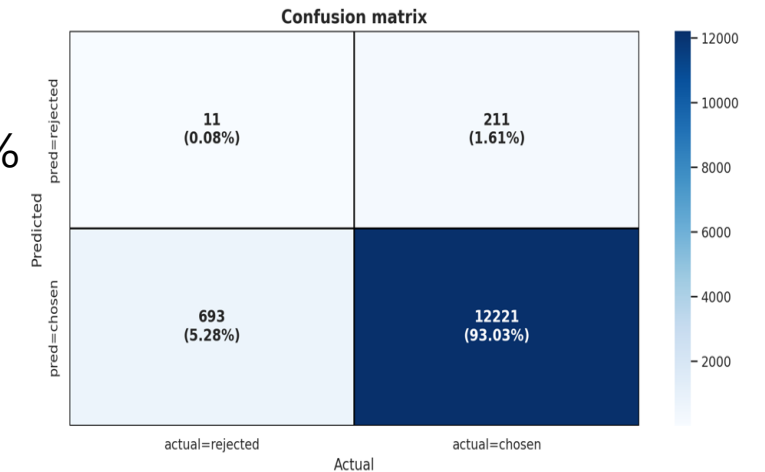
Accuracy: 98.74%

Weighted
Precision: 98.75%



Llama with DPO

Accuracy: 93.3%





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Demonstration

Demonstration

The screenshot shows a Google Colab notebook interface. The browser tabs at the top include 'UofT_PhishNet.ipynb - Colab', 'Phish Bowl Archives - Information Security at University of Toronto', and 'Gradio'. The notebook's menu bar includes 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. Below the menu, there are tabs for 'Commands', '+ Code', '+ Text', and 'Run all'. The left sidebar contains icons for file management and search. The main code cell shows the execution of `demo.launch(share=True, debug=True)`. The output indicates that the model was initialized on CUDA, some weights were not initialized, and the model was loaded successfully. It also provides a public URL for the demo: <https://5f17c20ce3b568bcf2.gradio.live>. Below the code output, a web interface for the 'UofT Phishing Detector' is displayed. The interface has a title 'UofT Phishing Detector' and a subtitle 'Upload an email PDF or paste text to analyze for phishing risks.' It features two input options: 'Paste Text' and 'Upload PDF'. The 'Paste Text' option is selected, showing a text area labeled 'Email Body' with the placeholder text 'Paste email content here...'. To the right of the text area is a 'Risk Assessment' section. At the bottom of the interface is a large blue button labeled 'Analyze Text'. The bottom status bar of the Colab interface shows 'Variables', 'Terminal', 'Executing (12m 37s)', and 'T4 (Python 3)'.

```
demo.launch(share=True, debug=True)
```

Initializing on cuda...
Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized. You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
✓ Model loaded successfully!
/tmp/ipython-input-732058854.py:126: DeprecationWarning: The 'theme' parameter in the Blocks constructor will be removed in Gradio 6.0. You will need to use with gr.Blocks(theme=gr.themes.Soft()) as demo:
Colab notebook detected. This cell will run indefinitely so that you can see errors and logs. To turn off, set debug=False in launch().
* Running on public URL: <https://5f17c20ce3b568bcf2.gradio.live>

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run ``gradio deploy`` from the terminal in the working directory to deploy your demo.

UofT Phishing Detector

Upload an email PDF or paste text to analyze for phishing risks.

[Paste Text](#) [Upload PDF](#)

Email Body

Paste email content here...

Risk Assessment

Analyze Text

Variables Terminal Executing (12m 37s) T4 (Python 3)

Discussion – Summary of Approaches

Model Name	Model Type	Phishnet Dataset Test Accuracy (%)	UofT Phishbowl Dataset Test Accuracy	Weighted Precision (%)	Num Parameters
RoBERTa	DL	98.75	64.29	98.75	124.6M
Llama 3.2 1B - DPO	DL	91.7- 98.5	35.9	*	1.2B
BERT	DL	98.47	64.29	98.47	108.3M
DistillBERT	DL	98.44	71.43	98.44	65.8M
Phish score	Hybrid	94	63.89	93.35	22.7M
Linear SVC (SVM)	ML	97.2	71.43	97.2	20.0K
Logistic Regression	ML	96.91	71.43	96.91	20.0K
SGD Classifier (SVM-like)	ML	96.72	35.71	96.72	20.0K
Naive Bayes (Baseline)	ML	94.75	85.71	94.82	40.0K



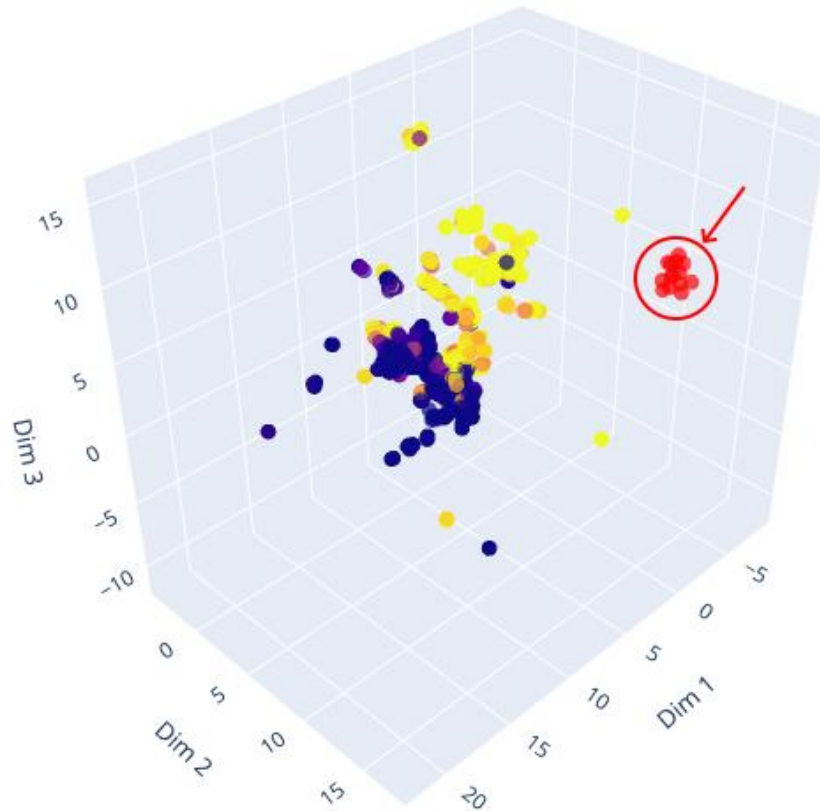
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Discussion

Discussion – Demonstration data in latent space

- Demonstration emails are out of distribution



2404 clusters colored by risk score and 14 colored in red

Labels

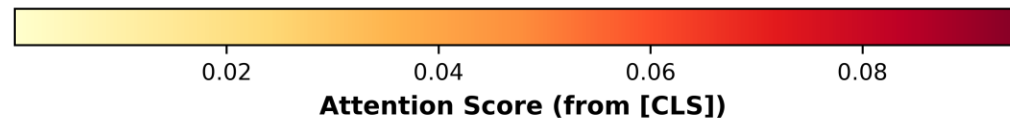
- Clusters, Blue = Great Yellow = Bait
- Demo Emails Embeddings

Discussion – Model Interpretability

- Attention scores for emails predicted **correctly** – Phishbowl dataset

Sample 2 | Pred: Bait

Title: Tuition scamDear Student,We are writing to remind you that our records indicate your Fall 2025 tuition fees remain unpaid. As per the University of Toronto's Financial Aid & Registration Policy, an initial deposit of 2,000 – 5,000 is required by today, August 31, 2025, in order to secure your course registration and maintain uninterrupted access to academic services.To complete your payment, please send an Interac e-Transfer to the University's BACPURSE account using the following details:Email: kellyvander9@outlook.comMessage/

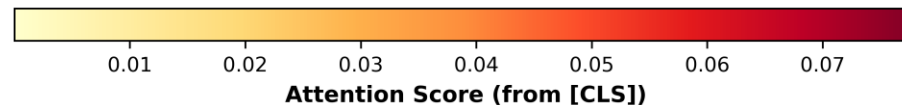


Discussion – Model Interpretability

- Attention scores for emails predicted **incorrectly** – Phishbowl dataset

Sample 7 | Pred: Great

Title: Get Ready to Take Your Research to the Next LevelHello,The Career Centre has shared with me some opportunities for jobs and conferences that may interest you.? My name is xxx and I am the Manager of Campus & Early Talent Recruitment for University Jobs Placement & Services, I'm reaching out to inform you of an exciting opportunity for students to provide feedback on our retail experience. We're seeking?students to participate in a paid evaluation program, sharing their honest opinions on our stores.Interns will work remotely, with flexible schedules accommodating up to 5-10 hours per week.

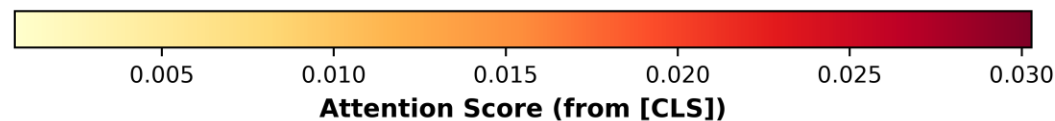


Discussion – Model Interpretability

- Attention scores for emails predicted **correctly** – Synthetic dataset

Sample 4 | Source: Emails_generated_by_Abhay | Pred: Bait (100.0%)

PayPal Account Limited - Suspicious Activity Detected Urgent Account Notification: We've identified potentially unauthorized access to your PayPal account and have implemented temporary security restrictions to protect your financial information. To restore full account functionality and verify your identity, you must immediately complete our security verification process at <http://paypal-account-verification-required.com/security-portal>. This essential security check requires you to confirm recent transactions, update your password, and review your account settings. The verification process typically takes 5-7 minutes and is mandatory to prevent fraudulent activity on your account. Delayed action may result in permanent account limitation and financial loss. For verification assistance, contact PayPal Security Support at security-verification@paypal-portal.com. Respectfully, PayPal Security Department



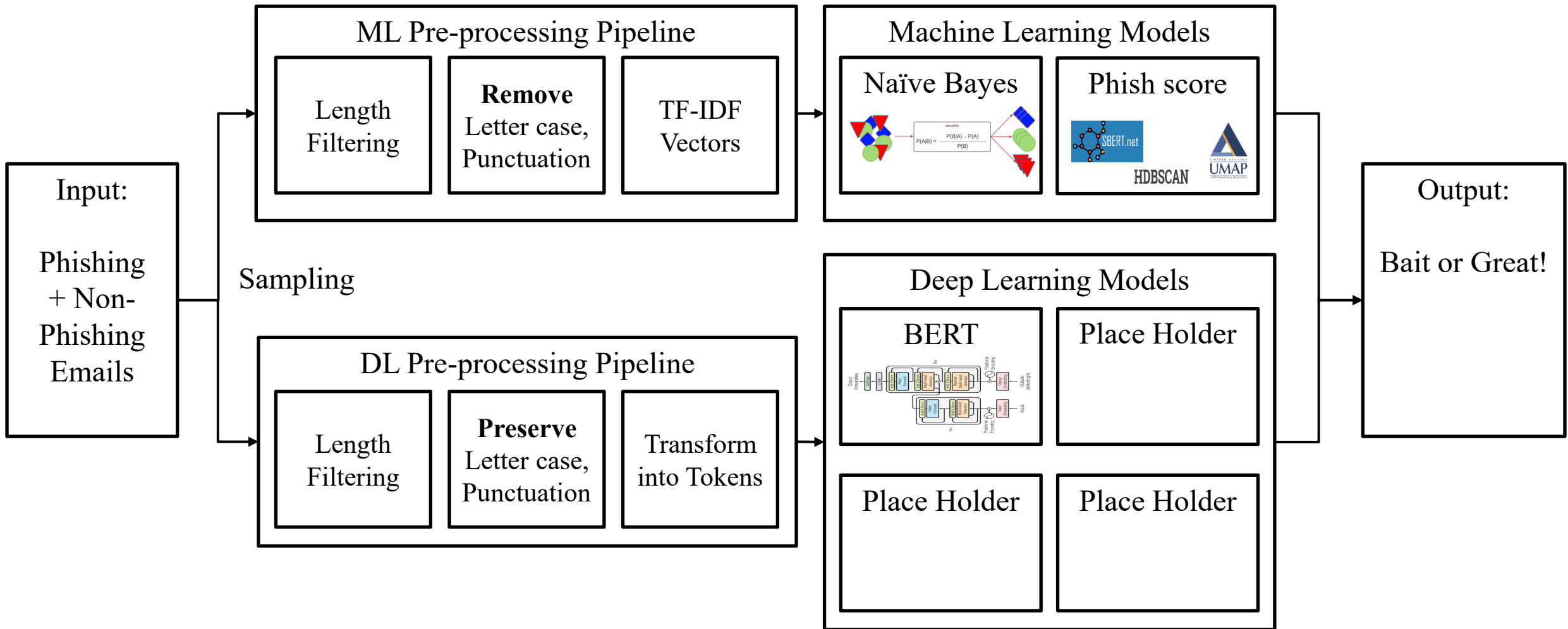


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Thank you.

End-to-End Pipeline



End-to-End Pipeline

